What changes without AF-Cluster?

Fresh Primary and HDBSCAN clustering on Mimas CPUs. The 11,104 retained models use the same coordinates, residue masks and score inputs as the original cohort.

Native ensemble DP rose by **+0.137 on average** for panels drawn from all eligible models (+0.05 to +0.20 across 36 matched comparisons). TM80 S1 panels changed by -0.002 on average (-0.03 to +0.02).

Fresh clustering on the three saved masks

Residues	Method	Clusters old / new	Noise old / new	ARI all	ARI assigned
1-139	Primary	25 / 15	0 / 0	0.424	0.424
1-139	HDBSCAN	142 / 127	5,763 / 5,967	0.536	0.919
5-95	Primary	27 / 21	0 / 0	0.965	0.965
5-95	HDBSCAN	142 / 128	5,510 / 5,173	0.647	0.953
32-47 + 78-99	Primary	33 / 19	0 / 0	0.090	0.090
32-47 + 78-99	HDBSCAN	139 / 129	5,436 / 5,215	0.338	0.897

Old counts are evaluated on the retained models. ARI is invariant to label names. The assigned-only column uses models assigned in both fits. Primary uses component-validated UMAP KNN. HDBSCAN refines the largest leaf in 10D UMAP.

Fresh native TM80 states

Residues	New state sizes	Unassigned	Old S1 retained	S1 Jaccard
1-139	10,922	182	10,922	1.00
5-95	10,920	184	10,920	1.00
32-47 + 78-99	920 / 870 / 133	9,181	1,520	0.45

Each state requires at least 50 models with bidirectional native TM at least 0.80 to its representative. The first two masks use TM core 15-72. The discontinuous mask uses core 32-47. State labels are local to each cohort. Jaccard compares the old and new S1 memberships on retained models.

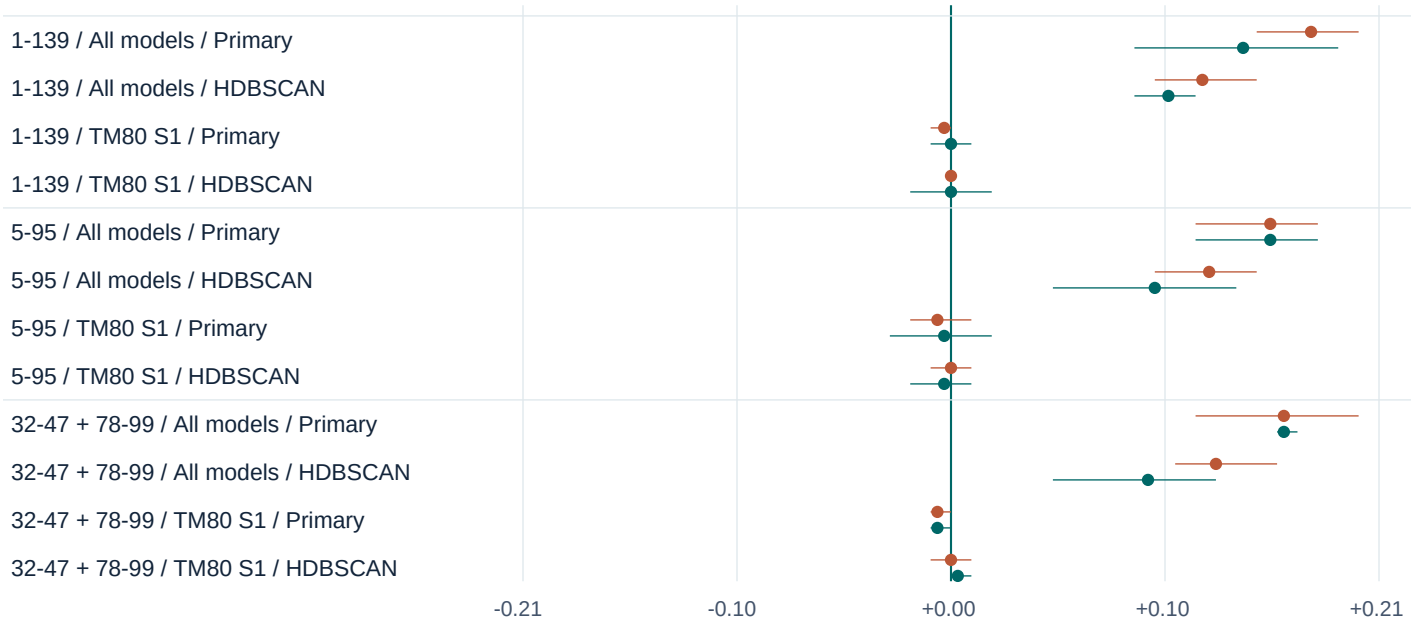
The original secondary states for 1-139 (85 and 133 models) and 5-95 (81 models) consisted entirely of AF-Cluster models. The new searches for those masks reached the candidate trial cap. The discontinuous-mask search reached the three-state limit.

MCMC50 and Gumbel50 after exclusion

The same three seeds were used for both cohorts. Each method selected 50 models in every residue-mask, clustering-prior and eligible-pool context.

On the filtered cohort, MCMC50 minus Gumbel50 averaged **+0.001 DP** (16 MCMC50 wins, 6 Gumbel50 wins, 14 ties). The original cohort gap was +0.009. Ties use native printed precision.

Native ensemble DP change after removing AF-Cluster



● MCMC50 ● Gumbel50 Filtered minus original DP. Dot = mean. Line = seed range.

Selector	Original mean	Filtered mean	Mean DP shift	Range of shifts
MCMC50	0.679	0.742	+0.063	-0.03 to +0.19
Gumbel50	0.669	0.741	+0.072	-0.02 to +0.20

Native RPF used the same original control and SR215-TR-2026-11-06-v1. The follow-up contains 72 panels and 72 new native calls. The cohort contrast includes reclustering, fresh state assignment and reweighting of the sampler. Recall, precision, mean member RPF, MolProbity and ProteinEBM are in the companion score table.